

CO2_ASSESSMENT TOOL_1

ANALYTICAL PROBLEM SOLVING

CSA0311-DS

CO2 - AT-1
Analytical Problem Solving.

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1. Convert the Infix Expression to postfix

Infix Expression

Step-by-Step:

$C/D \rightarrow CD/$
 $E \times F \rightarrow EF \times$
 $(E \times F) * G \rightarrow EF \times G *$
 $((E \times F) * G) / H \rightarrow EF \times G * H /$
 $C/D + ((E \times F) * G) / H \rightarrow CD/EF \times G * H / +$
Multiply by J : $CD/EF \times G * H / + J *$
Subtract from B

Postfix Expression

$CD/EF \times G * H / + J * -$

Queue Operation (Size 5).

Operations:

Queue

EnQueue(21)	21
enqueue (89)	21, 67, 89
enqueue (67)	21, 67, 89
Dequeue (1)	67, 89
enqueue (15)	67, 89, 15
enqueue (30)	67, 89, 15, 30
enqueue (52)	67, 89, 15, 30, 52
Dequeue (1)	89, 15, 30, 52
Dequeue (1)	15, 30, 52
Dequeue (1)	30, 52
Dequeue (1)	52

Final Queue : 52.

Conclusion,

The Queue operation were performed using the FIFO Principle. Element were inserted using enqueue and removed using Dequeue in order they were added. After programming all the operation Answer is 52.

Evaluate the postfix expressions

(a) $9\ 3\ 8\ 4\ |\ * +$

Step	Stack
9	9
3	9, 3
8	9, 3, 8
4	9, 3, 8, 4
/	9, 3, 2
*	9, 6
+	15

Answer: 15

b) $6\ 2\ +\ 6\ 3\ /$

$$6 + 2 = 8$$

$$6 / 3 = 2$$

Final Stack = 8, 2

Answer,

Invalid postfix expression (two values remain on the stack, so the expression is incomplete).